SWE 4740 **Embedded Software Development Lab** **Credit: 0.75**

Sessional works based on SWE 4739.

SWE 4738 **Computer Animation Lab** **Credit: 0.75**

Sessional works based on SWE 4737.

SWE 4742 **Computer and Information Security Lab** **Credit: 0.75**

Sessional works based on SWE 4741.

Eighth Semester

CSE 4809 **Algorithm Engineering** **Credit: 2.00**

Introduction and review of asymptotic analysis including big-oh notation, divide and conquer algorithms and its application in sorting, matrix multiplication etc., Median finding and selection, interval scheduling, the substitution method, the master method.

Introduction and applications of probability and randomized algorithms, quicksort and its analysis, radix sort, sorting lower bound, hashing, open addressing and amortization, amortized analysis.

The greedy algorithm design paradigms and its applications, dynamic programming design paradigm and its applications.

Graph primitives, BFS, DFS, topological sort in DAGS, all pairs shortest paths, minimum spanning trees and their applications to clustering, heaps and their applications.

Competitive analysis, network flow i.e. max flow and min cut algorithms, interlude: problem solving, van Emde Boas data structure.

Intractable problems and what to do about them, NP-completeness and the P vs. NP question, polynomial time approximations, sublinear-time algorithms, heuristics with provable performance guarantees, Approximation Algorithms, Fast Fourier Transform, local search, Linear Programming, exponential-time algorithms that beat brute-force search.

Recommended Texts:

1. T. H. Cormen et al., *Introduction to Algorithms*, 4th ed. The MIT press, 2022
2. A. Levitin, *Introduction to the Design and Analysis of Algorithms*, 3rd ed. Pearson, 2011

3. J. J. McConnell, *The Analysis of Algorithms: An Active Learning Approach*, 2nd ed. Jones & Bartlett Learning, 2007

SWE 4801 **Software Maintenance** **Credit: 3.00**

The nature of Software maintenance, Software Maintenance types;

Characteristics of maintainable software, Software Maintenance;

Process Models: Quick-and-Fix Model, Bohem's Model, Osborne Model, Iterative Model, State of the art tools for supporting software developers and maintenance engineers, Program Comprehension, Legacy Information Systems, Software Clone Detection and Analysis, Mining Software Repositories, Design Recovery, Traceability, Refactoring, Reuse and Domain Engineering, Locating features in source code, concept analysis, Program Transformation and Migration, Software Evolution Process Models, Software Testing during Maintenance and Evolution, Software Metrics for Maintenance;

Software Reuse, Maintenance and Evolution of Services Systems, Maintenance Cost Estimation by COCOMO II, Bohem's Maintenance Cost Model Impact Analysis, Big Data Analytics, Reverse engineering.

Recommended Texts:

1. S. Jarzabek, *Effective Software Maintenance and Evolution: A Reuse-Based Approach*, 1st ed. Auerbach Publications, 2007
2. P. Grubb and A. A. Takang, *Software Maintenance: Concepts and Practice*, 2nd ed. World Scientific Publishing Company, 2003

SWE 4803 **Software Project Management** **Credit: 3.00**

Project Management Basics, Role of a Project Manager, Project Resources, Phases of Software Project; Introduction to PERT/CPM, Software Project Planning: Management, Risk, Configuration, Quality Assurance, Induction, Schedule, Cost Estimation; People and Project Organization; Change Management Monitoring and Control; Productivity Aspects: Productivity Basics, Productivity Measurement & Metrics; Human Factors and Leadership: Motivation, Communication, Handling Difficult People, Leadership, Team Dynamics; Progress Tracking & Control: Progress Assessment & Reporting, Scope Management; Organizational Support for Effective Project Management; dispute and error tracking, RMMM charts Industry scenarios: Domain analysis, Business case analysis, Dynamicity, Success and failure factors, case studies.

Recommended Texts:

1. A. Stellman and J. Greene, *Applied Software Project Management*, 1st ed. O'Reilly Media, 2005